



ALLIANCE
UNIVERSITY

NAAC
GRADE A+
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ASSIGNMENT- 01

Bachelor of Computer Applications (Hons. Game Development)
Semester- III

Subject: Database Management System

Topic: Identifying Tables for a Real-World Application

By

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Library Management System

1. Name and Purpose of the Application

Name of the Application: Library Management System

Purpose: A Library Management System helps libraries (school, college, or public) manage their book collection, track who borrowed which books, handle reservations for unavailable books, calculate fines for overdue returns, and maintain library staff records. The system replaces manual registers with a structured database that ensures accuracy, prevents duplication, and allows quick retrieval of information such as book availability, borrowing history, and pending fines.

2. List of Required Tables

1. **Members** — library users who borrow books
2. **Staff**— library staffs
3. **Publishers** — companies that publish books
4. **Authors** — writers of books
5. **Categories** — genre/subject classification of books
6. **Books** — catalog of book titles held by the library
7. **BookCopies** — individual physical copies of each book title
8. **Book_Author** — junction table linking books to authors (many-to-many)
9. **Loans** — records of borrowing and returning transactions
10. **Reservations** — records of members reserving unavailable books
11. **Fines** — penalty records for late returns or damage

3. Columns/attributes

I. Members Table

Column	Type	Description
member_id	INT	(Primary Key) Gives a unique ID for each member
name	VARCHAR	Stores the name of the member
email	VARCHAR	Stores the email of the member
phone	VARCHAR	Stores the phone number of the member
address	VARCHAR	Stores the address of the member
membership_date	DATE	Stores the date joined
membership_status	VARCHAR	Stores the status (Active/Suspended/Expired)

Purpose: Stores personal and membership details of everyone registered to borrow books.

II. Staff Table

Column	Type	Description
staff_id	INT	(Primary Key) Gives a unique ID for each staff
name	VARCHAR	Stores the name of the staff
email	VARCHAR	Stores the email of the staff
phone	VARCHAR	Stores the phone number of the staff
designation	VARCHAR	Stores the designation of the staff
date_joined	DATE	Stores the date joined

Purpose: Maintains records of employees responsible for issuing books, managing fines, and overall library operations.

III. Publishers Table

Column	Type	Description
publisher_id	INT	(Primary Key) Gives a unique ID for each publisher
name	VARCHAR	Stores the name of the publisher
phone	VARCHAR	Stores the phone number of the publisher
address	VARCHAR	Stores the address of the publisher

Purpose: Stores details of publishing houses that produce the books in the library's collection.

IV. Authors Table

Column	Type	Description
author_id	INT	(Primary Key) Gives a unique ID for each author
name	VARCHAR	Stores the name of the author
nationality	VARCHAR	Stores the nationality of the author
date_of_birth	DATE	Stores the DOB

Purpose: Maintains a master list of authors, independent of any single book, since one author can write many books.

V. Categories Table

Column	Type	Description
category_id	INT	(Primary Key) Gives a unique ID for each category
category_name	VARCHAR	Stores the name of the category
description	VARCHAR	Stores the description of the category

Purpose: Classifies books by subject/genre for easier searching and organization.

VI. Books Table

Column	Type	Description
book_id	INT	Unique ID for each book
title	VARCHAR	Title of the book
isbn	VARCHAR	International Standard Book Number
publisher_id	INT	References Publishers
category_id	INT	References Category
publication_year	INT	Year of publication
language	VARCHAR	Language of the book

Purpose: Acts as the master catalog of all distinct book titles the library holds, regardless of how many physical copies exist.

VII. BookCopies Table

Column	Type	Description
copy_id	INT	Unique ID for each physical copy
book_id	INT	References book
rack_number	VARCHAR	Physical shelf/rack location
status	VARCHAR	Available/Issued/Damaged/Lost

Purpose: Tracks each physical copy of a book separately, since a popular title may have multiple copies with different availability statuses.

VIII. Book_Author (Junction Table)

Column	Type	Description
book_id	INT	References book
author_id	INT	References authors

Purpose: Resolves the many-to-many relationship between books and authors — a book can have multiple authors, and an author can write multiple books.

IX. Loan Table

Column	Type	Description
loan_id	INT	Unique ID for each loan transaction
copy_id	INT	References Book Copies
member_id	INT	References Members
staff_id	INT	References Staff
issue_date	DATE	Date the book was borrowed
due_date	DATE	Date by which it must be returned
return_date	DATE	Returned date

Purpose: The core transactional table recording every instance of a book copy being borrowed and returned

X. Reservations Table

Column	Type	Description
reservation_id	INT	Unique ID for each reservation
book_id	INT	References book
member_id	INT	References members
reservation_date	DATE	Date the reservation was made
status	VARCHAR	Pending/Fulfilled/Cancelled

Purpose: Allows members to reserve a book title (not a specific copy) that is currently unavailable, so they are notified when a copy is returned.

XI. Fines Table

Column	Type	Description
fine_id	INT	Unique ID for each fine
loan_id	INT	References loans
amount	DECIMAL	Fine amount
reason	VARCHAR	Late return/Damage/Lost
paid_status	VARCHAR	Paid/Unpaid
payment_details	DATE	Date the fine was paid

Purpose: Tracks penalties charged to members, linked directly to the specific loan transaction that caused the fine.

4. Relationships Between Tables

Relationship	Type	Description
Publishers → Books	One-to-Many	One publisher publishes many books
Categories → Books	One-to-Many	One category classifies many books
Books ↔ Authors	Many-to-Many	A book can have multiple authors; an author can write multiple books
Books → BookCopies	One-to-Many	One book title can have many physical copies
BookCopies → Loans	One-to-Many	One copy can be issued multiple times over its life (but only once at a time)
Members → Loans	One-to-Many	One member can take multiple loans over time
Staff → Loans	One-to-Many	One staff member can issue many loans
Books → Reservations	One-to-Many	One book title can be reserved by multiple members
Members → Reservations	One-to-Many	One member can make multiple reservations
Loans → Fines	One-to-One	Each fine is tied to exactly one loan record

5. Summary

This design uses 11 tables, each serving a distinct, necessary purpose:

- Master/reference tables: Members, Staff, Publishers, Authors, Categories, Books
- Structural table: BookCopies (separates "title" from "physical item")
- Junction table: Book_Author (resolves many-to-many relationship)
- Transactional tables: Loans, Reservations, Fines

Every table connects logically through primary and foreign keys, avoiding repeated or unnecessary entities while fully supporting the core operations of a real library: cataloging, issuing, returning, reserving, and fining.